Hsuan-Chi Liu

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Education

National Taiwan University — Taipei, Taiwan

B.S. in Computer Science and Information Engineering; GPA: 3.94/4.00 (4.15/4.30) — *Sep 2022 – Jun 2026*

- **Courses:** Parallel Programming, Computer Architecture, Reinforcement Learning, Machine Learning
- **Research Interests:** Efficient Machine Learning, High-Performance Computing, Generative Modeling

Publications

[*] Equal contribution; [†] Corresponding author.

Learning Conditional Source Distribution via Flow Reversal for Temporal Flow Matching
Kuan-Hsun Tu[*], **Hsuan-Chi Liu**[*†], Jia-Wei Liao[*], Chiensheng Chiang, Cheng-Fu Chou, and Tsung-Wei Ke.
In submission, 2026.

TeCoP: Accelerating Diffusion Large Language Model Decoding via Tentative Context Pipelining
Jui-Chien Tsou[*], **Hsuan-Chi Liu**[*], Wei-Chin Wang[*], Yu-Syuan Xu, and Chun-Yi Lee.
In submission, 2026.

Boosting Small Object Tracking via Collaborative Detection Transformer
Bing-Cheng Chuang[*], Jia-Hua Lee[*], Pin-Tseng Kuo[*], I-Hsuan Chu[*], Yi-Shein Hsiao[*], Cheng-Han Wu[*], Po-Yi Wu[*], Jui-Chien Tsou[*], **Hsuan-Chi Liu**[*], Chun-Yi Lee, and Yuan-Fu Yang.
19th International Conference on Machine Vision and Applications (MVA), IEEE, 2025.

Research Experience

Efficient Reinforcement Learning | Embodied AI Lab, NTU — Sep 2026 – Present

- Identifying inefficiencies in GPU-based robotic simulation and exploring optimization strategies to accelerate reinforcement learning training.
- Investigating how effective exploration strategies can improve sample efficiency and skill transfer in sparse-reward robotic manipulation tasks.

Archi-CRAB: Knowledge-Graph Agents for CMS Analysis Support — Jul 2026 – Oct 2026

- Collaborated with MIT's Archi team and CERN CMS to evaluate Operational Knowledge Graph (OKG) retrieval for LLM-based CRAB support, comparing evidence retrieval and answer quality against keyword-based and embedding-based retrieval baselines.
- Developed a semantically clustered benchmark of **178 real-world support tickets** and 69 FAQs, incorporating gold evidence annotations to assess retrieval recall and answer correctness.

VLM-Guided Action Sampling for Efficient RL Exploration — Nov 2024 – Oct 2025

- Designed a collaborative VLM–VLA framework for robot learning, using vision-language guidance to sample high-level action candidates and improve RL exploration efficiency and performance.

Image Stitching & SIFT Acceleration (OpenMP + ROCm HIP) — Oct 2024 – Dec 2024

- Accelerated an image-stitching pipeline by parallelizing SIFT preprocessing with OpenMP and offloading Gaussian blur convolutions to GPUs using ROCm HIP; integrated the optimized pipeline into OpenCV.

GPU Cluster Network Deployment (OPNsense HA Firewall/DNS) — Mar 2025 – Jun 2025

- Responsible for the firewall component: configured OPNsense for firewall and internal DNS, and set up CARP-based high availability for failover.

Awards & Competitions

Overall Champion, Student Cluster Competition (SCC25) Nov 2025
ACM/IEEE Supercomputing Conference *St. Louis, USA*

- Ran MLPerf Inference (Llama 3.1 8B) on 2-node H200 GPUs via Triton, achieving highest throughput among all teams under a 10 kW power cap.
- Accelerated GEMM in the Exascale Climate Emulator using cuBLASLt and mixed-precision techniques.

Second Place & Best AI Performance Award Nov 2025
APAC HPC-AI Competition, Asia Pacific Advanced Network *Singapore*

- Optimized MPI parallelization for the NWChem benchmark to reduce runtime and improve scalability.

Third Place, SMOT4SB Challenge (MVA2025) May 2025
19th International Conference on Machine Vision and Applications (MVA) *Kyoto, Japan*

First Place & Group Competition Award May 2025
ASC Student Supercomputer Challenge, Asia Supercomputer Community *Qinghai, China*

Work Experience

CERN Summer Student (CMS Offline & Computing) Jun 2026 – Aug 2026
European Organization for Nuclear Research (CERN) *Geneva, Switzerland*

- Developed MCP-based diagnostic tools to ground LLM agents in real-time CMS computing data, enabling evidence-based reasoning and automated troubleshooting.

Summer Research Intern (Streaming Graph Algorithms) Jun 2023 – Aug 2023
Institute of Information Science, Academia Sinica *Taipei, Taiwan*

- Explored approximation algorithms for minimum cycle cover in streaming graphs, focusing on space and computational efficiency.

Wireless Team Lead Aug 2025 – Present
Network Administration Team, NTU CSIE *Taipei, Taiwan*

- Administered the department-wide wireless network and collected logs for monitoring and troubleshooting.

Teaching Assistant (Parallel Programming) Sep 2025 – Dec 2025
National Taiwan University *Taipei, Taiwan*

Open-Source Contributions

MLCommons – MLPerf Inference 5.0 Nov 2025

- Contributed multi-node LLM inference configurations and documentation using NVIDIA Triton Inference Server for distributed MLPerf benchmarking.

Skills

Languages: Python, C/C++
GPU Acceleration: CUDA, cuBLAS/cuBLASLt, ROCm HIP
Parallel Computing: MPI, OpenMP
Profiling & Benchmarking: Nsight Systems, MLPerf Inference, multi-node scaling & tuning
Systems & Tools: Linux, Git, Docker, Slurm, OPNsense, DNS, LLM Agents, MCP, OpenSearch